

## Problem setting

Consider  $d = 1$ . For any policy  $\pi$ , let  $\mu^\pi = \text{law}(Z^\pi)$  be the law (distribution) of the return r.v.  $Z^\pi$  as defined in Eq. (1). The distributional Bellman operator  $\mathcal{T}^\pi$  (Bellemare, Dabney, and Munos 2017) specifies the relation of different return distributions across state-action pairs along the Bellman dynamic; that is, for any  $\mu \in \mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}}$ , and any  $(s, a) \in \mathcal{S} \times \mathcal{A}$ ,

$$\mathcal{T}^\pi \mu(x, a) := \int_{\mathcal{S}} \int_{\mathcal{A}} \int_{\mathcal{X}} (f_{\gamma, r})_{\#} \mu(s', a') \mathcal{R}(dr|s, a) \pi(da'|s') P(ds'|s, a),$$

where  $f_{\gamma, r}(z) := r + \gamma z, \forall z$  and  $(f_{\gamma, r})_{\#} \mu(s', a')$  is the pushforward measure of  $\mu(s', a')$  by  $f_{\gamma, r}$ . Note that  $\mu^\pi$  is the fixed point of  $\mathcal{T}^\pi$ , i.e.,  $\mathcal{T}^\pi \mu^\pi = \mu^\pi$ . We are interested in the problem of learning  $\mu^\pi \in \mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}}$  via the distributional Bellman operator  $\mathcal{T}^\pi$ .

## Algorithmic approach

In practical setting, we must approximate the return distribution via a finite set of statistics as the space of Borel probability measures is infinite-dimensional. Let  $Z_\theta(s, a) := \{Z_\theta(s, a)_i\}_{i=1}^N$  be a set of parameterized statistics of  $\mu^\pi(s, a)$  where  $\theta$  represents the parameters of the model, e.g., neural networks. Instead of restricting  $Z_\theta(s, a)$  to predefined statistic functionals, we model unrestricted statistics, i.e., deterministic samples where each  $Z_\theta(s, a)_i$  can be evolved into any form of statistics and we use the Dirac mixture  $\hat{\mu}_\theta(s, a) = \frac{1}{N} \sum_{i=1}^N \delta_{Z_\theta(s, a)_i}$  to approximate  $\mu^\pi(s, a)$ . We refer to the deterministic samples  $Z_\theta(s, a)$  as particles, and our goal is reduced into learning the particles  $Z_\theta(s, a)$  to approximate  $\mu^\pi(s, a)$ . To this end, the particles  $Z_\theta(s, a)$  is deterministically evolved to minimize the MMD distance between the approximate distribution and its distributional Bellman target. Algorithm 1 below presents the generic update in our approach, namely MMDRL.

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### Algorithm 1: Generic MMDRL update

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**Require:** Number of particles  $N$ , kernel  $k$ , discount factor  $\gamma \in [0, 1]$

**Input:** Sample transition  $(s, a, r, s')$

1 **if** Policy evaluation **then**

2    $a^* \sim \pi(\cdot|s')$

3 **end**

4 **else if** Control setting **then**

5    $a^* \leftarrow \arg \max_{a' \in \mathcal{A}} \frac{1}{N} \sum_{i=1}^N Z_\theta(s', a')_i$

6 **end**

7  $\hat{T}Z_i \leftarrow r + \gamma Z_{\theta-}(s', a^*)_i, \forall 1 \leq i \leq N$

**Output:**  $\text{MMD}_b^2 \left( \{Z_\theta(s, a)_i\}_{i=1}^N, \{\hat{T}Z_i\}_{i=1}^N; k \right)$

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**Intuition.** The MMDRL reduces into the standard TD or Q-learning when  $N = 1$ . For  $N > 1$ , the objective  $\text{MMD}_b$  when used with SGD and Gaussian kernels  $k(x, y) = \exp(-|x - y|^2/h)$  contributes in two ways: (i) The term  $\frac{1}{N^2} \sum_{i,j} k(Z_\theta(s, a)_i, Z_\theta(s, a)_j)$  serves as a repulsive force

that pushes the particles  $\{Z_\theta(s, a)_i\}$  away from each other, preventing them from collapsing into a single mode, with force proportional to  $\frac{2}{h} e^{-(Z_\theta(s, a)_i - Z_\theta(s, a)_j)^2/h} |Z_\theta(s, a)_i - Z_\theta(s, a)_j|$ ; (ii) the term  $-\frac{2}{N^2} \sum_{i,j} k(Z_\theta(s, a)_i, \hat{T}Z_j)$  acts as an attractive force which pulls the particles  $\{Z_\theta(s, a)_i\}$  closer to their target particles  $\{\hat{T}Z_j\}$ . This can also be intuitively viewed as a two-sample counterpart to Stein point variational inference (Liu and Wang 2016; Chen et al. 2018).

**Particle representation.** We can easily extend MMDRL to DQN-like architecture to create a novel deep RL, namely MMDQN (Algorithm 3 in Appendix B). In this work, we explicitly represent the particles  $\{Z_\theta(s, a)_i\}_{i=1}^N$  in MMDQN via fixed  $N$  network outputs as in QR-DQN (Dabney et al. 2018b) for simplicity (details in Appendix B). We emphasize that modeling improvements from IQN (Dabney et al. 2018a) and FQF (Yang et al. 2019) can be naturally applied to MMDQN: we can implicitly generate  $\{Z_\theta(s, a)_i\}_{i=1}^N$  via applying a neural network function to  $N$  samples of a base sampling distribution (e.g., normal or uniform distribution) as in IQN, or we can use the proposal network in FQF to learn the weights of each Dirac components in MMDQN instead of using equal weights  $1/N$ .

## Theoretical Analysis

Here we provide theoretical understanding of MMDRL. Before that, we define the notion of supremum MMD, a MMD counterpart to the supremum Wasserstein in (Bellemare, Dabney, and Munos 2017), to work on  $\mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}}$ .

**Definition 1.** Supremum MMD is a functional  $\mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}} \times \mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}} \rightarrow \mathbb{R}$  defined by

$$\text{MMD}_\infty(\mu, \nu; k) := \sup_{(x, a) \in \mathcal{S} \times \mathcal{A}} \text{MMD}(\mu(x, a), \nu(x, a); k)$$

for any  $\mu, \nu \in \mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}}$ .

We are concerned with the following questions:

1. **Metric property:** When does  $\text{MMD}_\infty$  induce a metric on  $\mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}}$ ?
2. **Contraction property:** When is  $\mathcal{T}^\pi$  a contraction in  $\text{MMD}_\infty$ ?
3. **Convergence property:** How fast do the particles returned by minimizing MMD approach the target distribution it approximates?

The metric property in the first question ensures that  $\text{MMD}_\infty$  is a meaningful test to distinguish two return distributions on  $\mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}}$ . The contraction property in the second question guarantees that following from Banach's fixed point theorem (Banach 1922),  $\mathcal{T}^\pi$  has a unique fixed point which is  $\mu^\pi$ . In addition, starting with an arbitrary point  $\mu_0 \in \mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}}$ ,  $\mathcal{T}^\pi \circ \mathcal{T}^\pi \circ \dots \circ \mathcal{T}^\pi \mu_0$  converges at an exponential rate to  $\mu^\pi$  in  $\text{MMD}_\infty$ . We provide sufficient conditions to answer the first two questions and derive the convergence rate of the optimal particles in approximating a target distribution for the third question. A short answer is that the first two properties highly depend on the underlying kernel  $k$ , and the particles returned by minimizing MMD enjoy a rate  $O(1/\sqrt{n})$  regardless of the dimension  $d$  of the underlying space  $\mathcal{X}$ .

## Metric property

**Proposition 1.** Let  $\tilde{\mathcal{P}}(\mathcal{S}) \subseteq \mathcal{P}(\mathcal{S})$  be some (Borel) subset of the space of the Borel probability measures. If MMD is a metric on  $\tilde{\mathcal{P}}(\mathcal{X})$ , then  $\text{MMD}_\infty$  is also a metric on  $\tilde{\mathcal{P}}(\mathcal{X})^{S \times A}$ .

*Proof.* See Appendix A.1.  $\square$

Theorem 1 below provides *sufficient conditions* for MMD to induce a metric on  $\tilde{\mathcal{P}}(\mathcal{X})$ .

**Theorem 1.** We have

1. If the underlying kernel  $k$  is **characteristic** (i.e., the induced Bochner integral  $\psi_p$  is injective), e.g., Gaussian kernels, then MMD is a metric on  $\mathcal{P}(\mathcal{X})$  (Fukumizu et al. 2007; Gretton et al. 2012).
2. Define **unrectified** kernels  $k_\alpha(x, y) := -\|x - y\|^\alpha, \forall \alpha \in \mathbb{R}, \forall x, y \in \mathcal{X}$ . Then  $\text{MMD}(\cdot, \cdot; k_\alpha)$  is a metric on  $\mathcal{P}_\alpha(\mathcal{X})$  for all  $\alpha \in (0, 2)$  but not a metric for  $\alpha = 2$  (Székely 2003).
3. MMD associated with the so-called **exp-prod** kernel  $k(x, y) = \exp(\frac{xy}{\sigma^2})$  for any  $\sigma > 0$  is a metric on  $\mathcal{P}_*(\mathcal{X})$ .

*Proof.* See Appendix A.2.  $\square$

**Contraction property** We analyze the contraction of  $\mathcal{T}^\pi$  for several important classes of kernels. One such class is shift invariant and scale sensitive kernels. A kernel  $k(\cdot, \cdot)$  is said to be *shift invariant* if  $k(x + c, y + c) = k(x, y), \forall x, y, c \in \mathcal{X}$ ; it is said to be *scale sensitive* with order  $\alpha > 0$  if  $k(cx, cy) = |c|^\alpha k(x, y), \forall x, y \in \mathcal{X}$  and  $c \in \mathbb{R}$ . For example, the unrectified kernel  $k_\alpha$  considered in Theorem 1 is both shift invariant and scale sensitive with order  $\alpha$  while Gaussian kernels are only shift invariant.

**Theorem 2.** We have

1. If the underlying kernel is  $k = \sum_{i \in I} c_i k_i$  where each component kernel  $k_i$  is both shift invariant and scale sensitive with order  $\alpha_i > 0, c_i \geq 0$ , and  $I$  is a (possibly infinite) index set, then  $\mathcal{T}^\pi$  is a  $\gamma^{\alpha_*/2}$ -contraction in  $\text{MMD}_\infty$  where  $\alpha_* := \min_{i \in I} \alpha_i$ .
2.  $\mathcal{T}^\pi$  is **not** a contraction in  $\text{MMD}_\infty$  associated with either Gaussian kernels or exp-prod kernels  $k(x, y) = \exp(\frac{xy}{\sigma^2})$ .

*Proof.* See Appendix A.3.  $\square$

**Practical consideration.** Theorem 2 provides a negative result for the commonly used Gaussian kernel. In practice, however, we found that Gaussian kernels can promote to match the moments between two distributions and have better empirical performance as compared to the other kernels analyzed in this section. In fact, MMD associated with Gaussian kernels  $k(x, y) = \exp(-(x - y)^2/(2\sigma^2))$  can be decomposed into

$$\text{MMD}^2(\mu, \nu; k) = \sum_{n=0}^{\infty} \frac{1}{\sigma^{2n} n!} (\tilde{m}_n(\mu) - \tilde{m}_n(\nu))^2$$

where  $\tilde{m}_n(\mu) = \mathbb{E}_{x \sim \mu} [e^{-x^2/(2\sigma^2)} x^n]$ , and similarly for  $\tilde{m}_n(\nu)$ . This indicates that MMD associated with Gaussian

kernels approximately performs moment matching (scaled with a factor  $e^{-x^2/(2\sigma^2)}$  for each moment term).

**Convergence rate of distribution approximation** We justify the goodness of the particles obtained via minimizing MMD in terms of approximating a target distribution.

**Theorem 3.** Let  $\mathcal{X} \subseteq \mathbb{R}^d$  and  $P \in \mathcal{P}(\mathcal{X})$ . For any  $n \in \mathbb{N}$ , let  $\{x_i\}_{i=1}^n \subset \mathcal{X}$  be a set of  $n$  deterministic points such that  $\{x_i\}_{i=1}^n \in \arg \inf_{\{\tilde{x}_i\}_{i=1}^n} \text{MMD}(\frac{1}{n} \sum_{i=1}^n \delta_{\tilde{x}_i}, P; \mathcal{F})$ . Then,  $P_n := \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$  converges to  $P$  at a rate of  $O(1/\sqrt{n})$  in the sense that for any function  $h$  in the unit ball of  $\mathcal{F}$ , we have

$$\left| \int_{\mathcal{X}} h(x) dP_n(x) - \int_{\mathcal{X}} h(x) dP(x) \right| = O(1/\sqrt{n}).$$

**Remark.** MMD enjoys a convergence rate of  $O(n^{-1/2})$ <sup>1</sup> regardless of the underlying dimension  $d$  while 1-Wasserstein distance has a convergence rate of  $O(n^{-1/d})$  (if  $d > 2$ ) (Fournier and Guillin 2015), which is slower for large  $d$ .

*Proof.* We first present two relevant results below (whose proofs are deferred to Appendix A.4) from which the theorem can follow.

**Proposition 2.** Let  $(X_i)_{i=1}^n$  be  $n$  i.i.d. samples of some distribution  $P$ . We have

$$\text{MMD} \left( \frac{1}{n} \sum_{i=1}^n \delta_{X_i}, P; \mathcal{F} \right) = O_p(1/\sqrt{n}),$$

where  $O_p$  denotes big- $O$  in probability.

**Lemma 1.** Let  $(a_n)_{n \in \mathbb{N}} \subset \mathbb{R}$  and  $(X_n)_{n \in \mathbb{N}} \subset \mathbb{R}$  be sequences of deterministic variables and of random variables, respectively, such that for all  $n, |a_n| \leq |X_n|$  almost surely (a.s.). Then, if  $X_n = O_p(f(n))$  for some function  $f(n) > 0$ , we have  $a_n = O(f(n))$ .

It follows from the Cauchy-Schwartz inequality in  $\mathcal{F}$  that for any function  $h$  in the unit ball of  $\mathcal{F}$ , we have

$$\begin{aligned} & \left| \int_{\mathcal{X}} h(x) dP_n(x) - \int_{\mathcal{X}} h(x) dP(x) \right| \\ & \leq \|h\|_{\mathcal{F}} \times \left\| \int_{\mathcal{X}} k(x, \cdot) dP_n(x) - \int_{\mathcal{X}} k(x, \cdot) dP(x) \right\|_{\mathcal{F}} \\ & \leq \text{MMD} \left( \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, P; \mathcal{F} \right). \end{aligned} \quad (2)$$

Now by letting  $X_n = \text{MMD}(\frac{1}{n} \sum_{i=1}^n \delta_{x_i}, P; \mathcal{F})$ ,  $a_n = \text{MMD}(\frac{1}{n} \sum_{i=1}^n \delta_{x_i}, P; \mathcal{F})$ , and  $f(n) = 1/\sqrt{n}$ , and noting that  $a_n \leq X_n, \forall n$ , Proposition 2, Lemma 1 and Eq. (2) immediately imply Theorem 3.  $\square$

<sup>1</sup>In fact, the convergence rate can be further improved to  $O(1/n)$  using kernel herding technique (Chen, Welling, and Smola 2010).

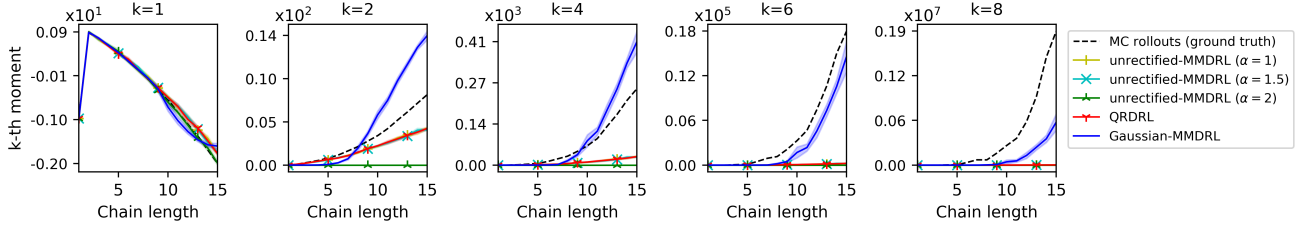


Figure 1: Performance of different methods in approximating the optimal policy’s return distribution at the initial state in the chain environment of various chain lengths  $K = \{1, 2, \dots, 15\}$ . The distribution approximation is evaluated in terms of how well a method can approximate the  $k$ -th central moment (except  $k = 1$  means the expectation) of the target distribution. 95% C.I. with 30 seeds. A variant (Gaussian-MMDRL) of our proposed MMDRL matches the MC rollouts (representing ground truth) much better than QRDL.

## Experimental Results

We first present results with a tabular version of MMDRL to illustrate its behaviour in distribution approximation task. We then combine the MMDRL update to the DQN-style architecture to create a novel deep RL algorithm namely MMDQN, and evaluate it on the Atari-57 games. We give full details of the architectures and hyperparameters used in the experiments in the Appendix B.<sup>2</sup>

### Tabular policy evaluation

We empirically evaluate that MMDRL with Gaussian kernels (*Gaussian-MMDRL*) can approximately learn the moments of a policy’s return distribution as compared to the MMDRL with unrectified kernels (*unrectified-MMDRL*) and the baseline QRDL.

We use a variant of the classic chain environment (Rowland et al. 2019). The chain environment of length  $K$  is a chain of  $K$  states  $s_0, \dots, s_{K-1}$  where  $s_0$  is the initial state and  $s_{K-1}$  is the terminal state (see Figure 2). In each state, there are only two possible actions: (i) forward, which moves the agent one step to the right with probability 0.9 and to  $s_0$  with probability 0.1, or (ii) backward, which transitions the agent to  $s_0$  with probability 0.9 and one step to the right with probability 0.1. The agent receives reward  $-1$  when transitioning to the initial state  $s_0$ , reward 1 when reaching the terminal state  $s_{K-1}$ , and 0 otherwise. The discount factor is  $\gamma = 0.9$ . We estimate  $\mu_0^*$  the return distribution at the initial state of the optimal policy  $\pi^*$  which selects forward action in every state. The longer the chain length  $K$ , the more stochastic the optimal policy’s return distribution at  $s_0$ . We use 10,000 Monte Carlo rollouts under policy  $\pi^*$  to compute the central moments of  $\mu_0^*$  as ground truth values. Each method uses only  $N = 30$  samples to approximate the target distribution  $\mu_0^*$  (more algorithm details are presented in Algorithm 2 in Appendix B).

The result is presented in Figure 1. While all the methods approximate the expectation of the target distribution well, their approximation qualities differentiate greatly when it comes to higher order moments. Gaussian-MMDRL, though with only  $N = 30$  particles, can approximate higher order

moments more reasonably in this example whereas the rest highly suffer from underestimation. We also experimented with the kernel considered in Theorem 1.3 in this tabular experiment and the Atari game experiment (next part) but found that it is highly inferior to the other kernel choices (even though it has an exact moment matching form as compared to Gaussian kernels, see Appendix A.2) thus we did not include it (we speculate that the shift invariance of Gaussian kernels seems effective when interacting with transition samples from the Bellman dynamics).

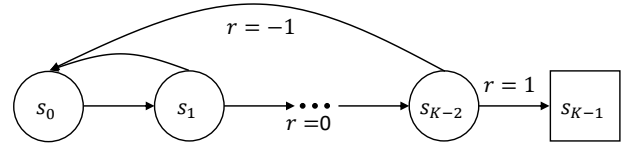


Figure 2: An illustration of a variant of the classic chain MDP with the chain length  $K$ .

### Atari games

To demonstrate the effectiveness of MMDRL at scale, we combine the MMDRL in Algorithm 1 with DQN-like architecture to obtain a deep RL agent namely MMDQN (Algorithm 3 in Appendix B). Specifically in this work, we used the same architecture as QR-DQN (Dabney et al. 2018b) for simplicity but more advanced modeling improvements from IQN (Dabney et al. 2018a) and FQF (Yang et al. 2019) can naturally be used in combination to our framework.

**Evaluation Protocol.** We evaluated our algorithm on 55<sup>3</sup> Atari 2600 games (Bellemare et al. 2013) following the standard training and evaluation procedures (Mnih et al. 2015; van Hasselt, Guez, and Silver 2016) (the full details are in appendix B). We computed human normalized scores for each agent per game. From the human normalized scores for an agent across all games, we extracted three statistics for the agent’s performance: the median, the mean and the number of games where the agent’s performance is above the human expert’s performance.

<sup>2</sup>The code for the experiments is available at <https://github.com/thanhnguyentang/mmdrl>.

<sup>3</sup>We failed to include Defender and Surround games using OpenAI and Dopamine framework.